

03_DIGITAL_TWIN_RESULTS

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EVIDENCE PROVENANCE: [SIMULATION / DETERMINISTIC]

Evaluated over multi-seed 15-step scenarios on digitized NMDC Bailadila Deposit 5 haulage topography.

1. Topographic Geodata Fidelity

The Digital Twin accurately reflects NMDC Bailadila Deposit 5 topography: 3.8 km haul route from RL 1180m to RL 920m, 8% nominal and 12% max downhill ramp grades, and 22m radius hairpin switchbacks.

2. Multi-Seed Deterministic Results

Benchmark Metric	Seed 0	Seed 42	Seed 100	Result
Twin Ingestion Rate	10.0 Hz	10.0 Hz	10.0 Hz	Rock-solid 10 Hz
Fog Ingestion Delay	14.2 ms	15.0 ms	14.8 ms	< 20.0 ms target
Governor Convergence	42.0 ms	43.5 ms	41.8 ms	< 50.0 ms target
Safety Violations (v > v_safe)	0 violations	0 violations	0 violations	100% Invariant
Switchback Max Queue	2 dumpers	2 dumpers	2 dumpers	Bounded queue
Throughput Retention	68.4%	68.2%	68.5%	Consistent